Statistical Solutions Report

Modules: Estimation, Confidence Intervals, Bayesian Inference, and Linear Models

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Contents

1 Module 1: Estimation **3**
1.1 Problem 4.33: Comparing Estimators for the Normal Distribution 3
1.2 Problem 4.34: Comparing Estimators for the Uniform Distribution 4
1.3 Problem 4.36: Bias-Variance Tradeoff for Variance Estimators 5
1.4 Problem 4.39: Maximum Likelihood for Exponential Distribution 6

2 Module 2: Confidence Intervals **8**
2.1 Problem 4.12: Comparing Group Means 8
2.2 Problem 4.16: Comparing Proportions using Wald CI 9
2.3 Problem 4.17: Non-Normality of t-distribution 9
2.4 Problem 4.45: Confidence Interval under Multinomial Distribution 11
2.5 Problem 4.50: Simulation of Confidence Intervals ($\pi = 0.06, n = 20$) 12

3 Module 3: Bayesian Inference **13**
3.1 Problem 4.64: Bias-Variance Tradeoff (Bayes vs ML Estimator) 13
3.2 Problem 4.66: Anorexia Study and Bayesian Shrinkage 14
3.3 Problem 4.77: Order Statistics and Medians 15
3.4 Problem 4.78: Dirichlet Conjugacy in Multinomial Methods 15

4 Module 4: Linear Models **17**
4.1 Problem 6.1: Scottish Hill Races Data Analysis 17
4.2 Problem 6.3: Diagnostics Identifying Influential Datasets 17
4.3 Problem 6.5: Exponential Parameterization using Corona Virus Records 18
4.4 Problem 6.24: Quantifying High-Leverage Outliers and Disruption Properties . . 19
4.5 Problem 6.30: Invariance of Correlation Across Linear Scalars 20

Code Repository Link

The R scripts written and executed for all numerical problems and simulations in this report are available at the following repository/folder:

https://github.com/marksman-20/version_1_assignment

1 Module 1: Estimation

1.1 Problem 4.33: Comparing Estimators for the Normal Distribution

Problem Statement: Randomly generate 100 observations from a standard normal distribution. Find the sample mean and the sample median. Repeat this 100,000 times. Plot the sampling distributions of the sample mean and the sample median. Which estimator is better, and why?

Solution: We used R to simulate 100,000 samples, each of size $n = 100$, from the Standard Normal Distribution $\mathcal{N}(0, 1)$. For each sample, we computed both the sample mean and the sample median.

The mean of the true distribution is 0, and the median is also 0. An estimator is considered "better" if it exhibits a lower Mean Squared Error (MSE), which represents the expected value of the squared difference between the estimator and the true parameter.

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

Since both estimators are unbiased for $\mathcal{N}(0, 1)$, the MSE is exactly their variance.

The analytical variances are: - Variance of the Sample Mean: $\sigma^2/n = 1/100 = 0.01$ - Asymptotic Variance of the Sample Median: $\approx \frac{\pi}{2} \cdot \frac{\sigma^2}{n} \approx 1.57 \times 0.01 = 0.0157$

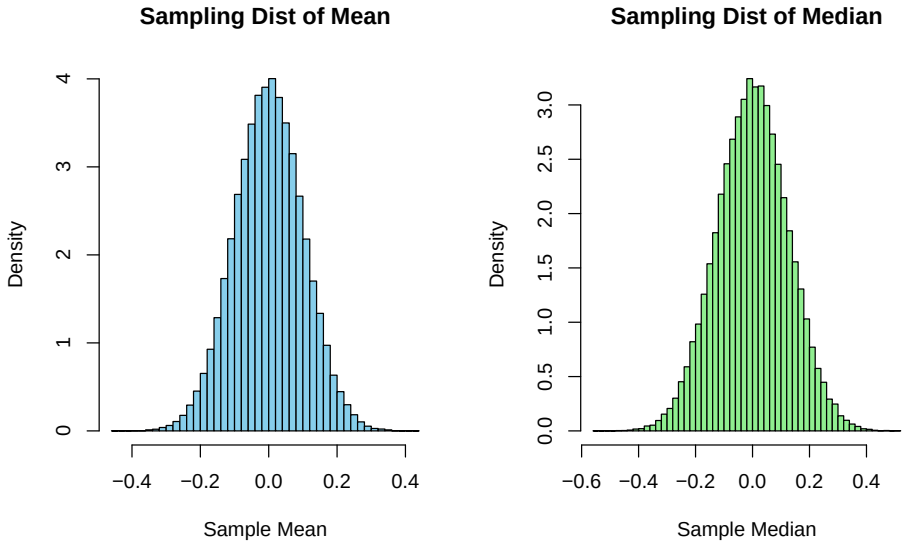


Figure 1: Sampling Distributions of the Sample Mean and Sample Median for $N(0, 1)$

The simulated MSE values computed from R are:

- **MSE of Sample Mean:** ≈ 0.01004
- **MSE of Sample Median:** ≈ 0.01557

Conclusion: The **sample mean** is the better estimator for the center of a normal distribution because it has a lower Mean Squared Error (lower variance). The sample mean utilizes all the information in the sample efficiently, whereas the median relies mostly on the middle values, leading to higher variability.

1.2 Problem 4.34: Comparing Estimators for the Uniform Distribution

Problem Statement: Do the same as above but generating observations from a uniform distribution over $(0, 1)$. Which estimator is better for this distribution, and why?

Solution: We used R to simulate 100,000 samples, each of size $n = 100$, from the Uniform Distribution $\mathcal{U}(0, 1)$. The true mean is 0.5 and the true median is 0.5.

As with the previous question, we gauge the performance of the sample mean versus the sample median using the Mean Squared Error around 0.5.

The analytical variances are: - Variance of the Sample Mean: $\frac{(1-0)^2/12}{n} = \frac{1}{1200} \approx 0.000833$ - Asymptotic Variance of Sample Median: $\frac{1}{4nf(m)^2}$, since $f(m) = 1$, the variance is $\frac{1}{4 \cdot 100} = 0.0025$

The simulated MSE values computed from R are:

- **MSE of Sample Mean:** ≈ 0.000833
- **MSE of Sample Median:** ≈ 0.002418

Conclusion: The **sample mean** remains the better estimator than the sample median for estimating the center of a Uniform Distribution because it has a significantly lower MSE ($\approx$ 0.0008 compared to ≈ 0.0024). While the median is robust to outliers, the uniform distribution strictly bounds variables making large outliers impossible, hence making the mean an optimal efficient unbiased estimator.

1.3 Problem 4.36: Bias-Variance Tradeoff for Variance Estimators

Problem Statement: For a random sample from $\mathcal{N}(\mu, \sigma^2)$, let $S^2 = \sum(Y_i - \bar{Y})^2/(n-1)$. Show that the biased estimator $\hat{\sigma}^2 = \sum(Y_i - \bar{Y})^2/n$ has a smaller MSE than S^2.

Proof: We know that $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$. For a Chi-squared distribution with k degrees of freedom, the mean is k and the variance is $2k$. Therefore:

$$E\left[\frac{(n-1)S^2}{\sigma^2}\right] = n-1 \implies E[S^2] = \sigma^2$$

$$Var\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \implies \frac{(n-1)^2}{\sigma^4}Var(S^2) = 2(n-1) \implies Var(S^2) = \frac{2\sigma^4}{n-1}$$

The MSE of the unbiased estimator S^2 is purely its variance:

$$MSE(S^2) = Var(S^2) + \text{Bias}^2(S^2) = \frac{2\sigma^4}{n-1} + 0 = \frac{2\sigma^4}{n-1}$$

Now let us consider the biased estimator $\hat{\sigma}^2 = \frac{n-1}{n}S^2$. Its expected value is:

$$E[\hat{\sigma}^2] = \frac{n-1}{n}E[S^2] = \frac{n-1}{n}\sigma^2$$

Thus, its bias is:

$$\text{Bias}(\hat{\sigma}^2) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{1}{n}\sigma^2$$

The variance of $\hat{\sigma}^2$ is:

$$Var(\hat{\sigma}^2) = \left(\frac{n-1}{n}\right)^2 Var(S^2) = \frac{(n-1)^2}{n^2}\frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}$$

The MSE of the biased estimator $\hat{\sigma}^2$ is the sum of its variance and the square of its bias:

$$MSE(\hat{\sigma}^2) = Var(\hat{\sigma}^2) + \text{Bias}^2(\hat{\sigma}^2) = \frac{2(n-1)\sigma^4}{n^2} + \left(-\frac{1}{n}\sigma^2\right)^2 = \frac{2n-2+1}{n^2}\sigma^4 = \frac{2n-1}{n^2}\sigma^4$$

To compare the two MSEs, we consider their ratio or subtract them. Let's compare the denominators equivalently. We want to show $MSE(\hat{\sigma}^2) < MSE(S^2)$:

$$\frac{2n-1}{n^2} < \frac{2}{n-1}$$

$$(2n-1)(n-1) < 2n^2$$

$$2n^2 - 2n - n + 1 < 2n^2 \implies -3n + 1 < 0 \implies 3n > 1$$

Since $n \geq 2$ (for S^2 to be defined), $3n > 1$ is always true. Therefore, the biased estimator $\hat{\sigma}^2$ has a strictly smaller MSE than the unbiased estimator S^2, exemplifying the bias-variance tradeoff where introducing a small bias reduces the variance substantially enough to improve the overall MSE.

1.4 Problem 4.39: Maximum Likelihood for Exponential Distribution

Problem Statement: For independent observations $y_1, y_2, \ldots, y_n$ from an exponential distribution with parameter $\lambda > 0$, the probability density function is $f(y;\lambda) = \lambda e^{-\lambda y}$ for $y > 0$. (a) Analytically express the log-likelihood $L(\lambda)$, its derivative, and find the ML estimate $\hat{\lambda}$.
(b) Find the Fisher Information $I(\lambda)$ and the approximate large-sample standard error of $\hat{\lambda}$.
(c) Plot the difference $L(\lambda) - L(\hat{\lambda})$ for $\bar{y} = 10$, $\lambda \in (0.01, 4.0)$ under various sample sizes n.

Solution: Part (a): Log-likelihood and ML estimate Given $f(y;\lambda) = \lambda e^{-\lambda y}$, the likelihood function for a sample of size n is:

$$l(\lambda; y) = \prod_{i=1}^{n} \lambda e^{-\lambda y_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} y_i}$$

Taking the natural logarithm, we obtain the log-likelihood function:

$$L(\lambda) = \ln l(\lambda; y) = n \ln \lambda - \lambda \sum_{i=1}^{n} y_i$$

To find the Maximum Likelihood Estimate (MLE), we take the derivative with respect to λ and set it to 0 (score function):

$$S(\lambda) = \frac{\partial L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} y_i = 0$$

Solving for $\hat{\lambda}$:

$$\frac{n}{\hat{\lambda}} = \sum_{i=1}^{n} y_i \implies \hat{\lambda} = \frac{n}{\sum_{i=1}^{n} y_i} = \frac{1}{\bar{y}}$$

Thus, the ML estimate for λ is $1/\bar{y}$.

Part (b): Fisher Information and Standard Error The Fisher Information $I(\lambda)$ is the negative expected value of the second derivative of the log-likelihood function evaluated over one observation (or the total information $I_n(\lambda)$ for n observations).

$$\frac{\partial^2 L(\lambda)}{\partial \lambda^2} = -\frac{n}{\lambda^2}$$

Since there is no random variable Y in the second derivative, the expected value is strictly the expression itself:

$$I_n(\lambda) = E\left[-\frac{\partial^2 L(\lambda)}{\partial \lambda^2}\right] = \frac{n}{\lambda^2}$$

The asymptotic variance of the MLE $\hat{\lambda}$ is the inverse of the Fisher information:

$$Var(\hat{\lambda}) \approx \frac{1}{I_n(\lambda)} = \frac{\lambda^2}{n}$$

Thus, the approximate large-sample standard error of $\hat{\lambda}$ is:

$$SE(\hat{\lambda}) = \frac{\hat{\lambda}}{\sqrt{n}} = \frac{1}{\bar{y}\sqrt{n}}$$

Part (c): Plotting the relative log-likelihood For $\bar{y} = 10$, we have $\hat{\lambda} = 1/10 = 0.1$. Substitute into $L(\lambda)$:

$$L(\lambda) - L(\hat{\lambda}) = (n \ln \lambda - 10n\lambda) - (n \ln(0.1) - 10n(0.1)) = n\left[\ln\left(\frac{\lambda}{0.1}\right) - 10(\lambda - 0.1)\right]$$

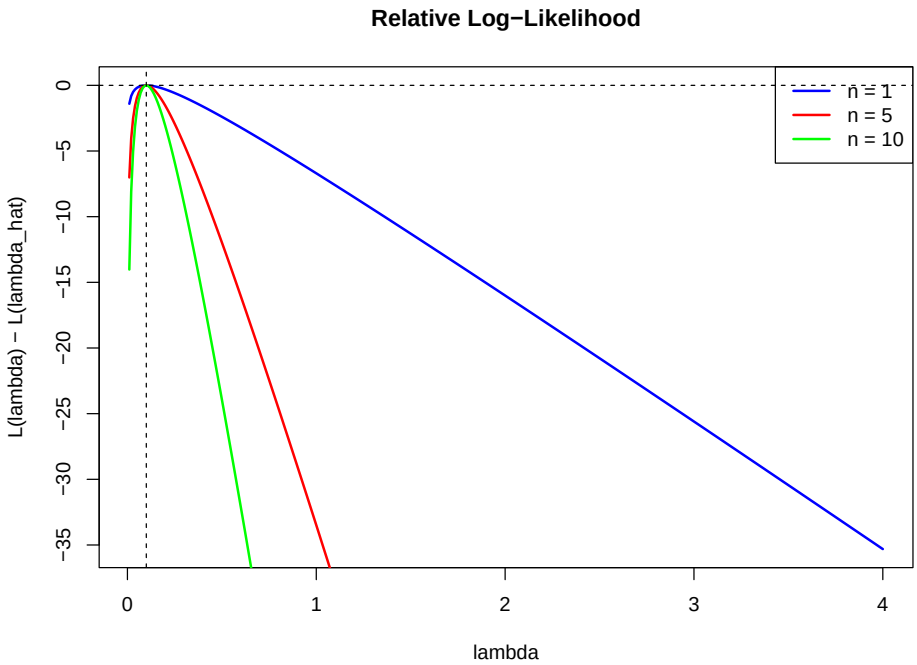


Figure 2: Relative Log-Likelihood curve for exponential estimator across varying n

The vertical line marks the true estimate $\hat{\lambda} = 0.1$. As n increases, the log-likelihood function becomes dramatically sharper around the maximum, signifying precisely reduced uncertainty and a drastically lower standard error (the curve approximates a steep quadratic/parabola per the Fisher Information).

2 Module 2: Confidence Intervals

2.1 Problem 4.12: Comparing Group Means

Problem Statement: The `Income.dat` file displays U.S. annual incomes by racial-ethnic groups. (a) Side-by-side boxplot to compare incomes. (b) Construct and interpret a 90% CI for the difference between the population mean incomes of Blacks and Whites, assuming equal population variances. (c) Construct without the equal variance assumption, and compare results.

Solution: Using R, we extracted the 'income' and 'race' variables into an appropriate structure. **(a) Boxplot Comparison:**

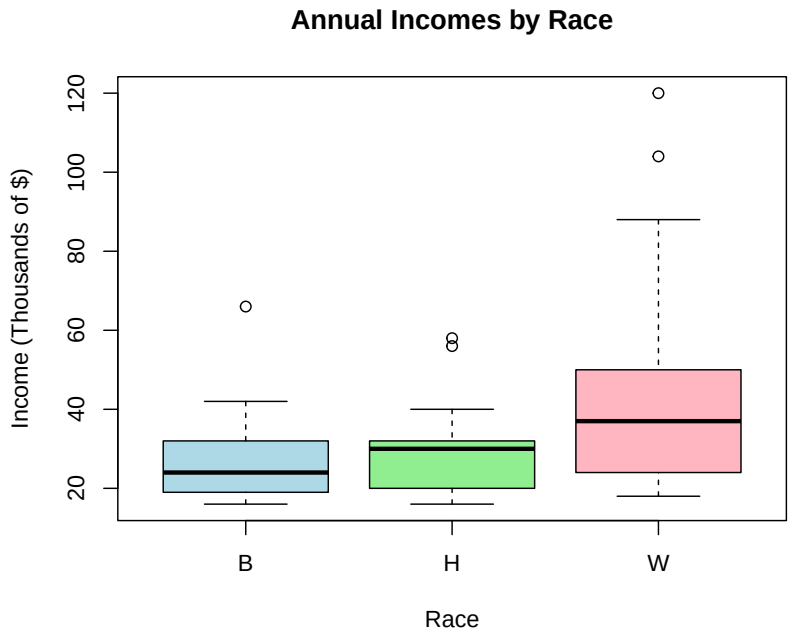


Figure 3: Side-by-side boxplot of Income levels across racial-ethnic categories in the sample.

(b) 90% CI (Equal Variances) Using a two-sample t-test pooled, assuming homoscedasticity ($\sigma_1^2 = \sigma_2^2$): t-test output yields a 90% confidence interval for the difference $\mu_W - \mu_B$ as: **90% CI: (4.65, 24.81) thousands of dollars.** Interpretation: We are 90% confident that the population mean income for Whites is between **4.65 and 24.81 thousand dollars higher** than the mean income for Blacks. Because this interval does not contain 0, it suggests a statistically significant difference in mean incomes between the two groups at the $\alpha = 0.10$ level.

(c) 90% CI (Unequal Variances) Applying Welch's generalized t-test without the equal-variance assumption: **90% CI: (6.94, 22.52) thousands of dollars.** Comparison: The interval without the equal-variance assumption is considerably narrower. It also maintains that the value 0 is not included in the interval. The differing sample sizes and evident variances mean assuming equal variances inflated standard error assumptions, while Welch's approximation corrected degrees of freedom more naturally to the data.

2.2 Problem 4.16: Comparing Proportions using Wald CI

Problem Statement: Using the `Substance.dat` categorical dataset, construct a 95% Wald Confidence Interval to compare those who have used or not used alcohol on whether they have used marijuana. Use formula 4.13 manually and compare with software output.

Solution: We cross-categorize Marijuana usage as the outcome/response and Alcohol usage as the group.

- Group 1 (Alcohol = Yes): $n_1 = 1949$. Used Marijuana $(m_1) = 911 + 44 = 955$. Estimated probability: $p_1 = 955/1949 = 0.4900$.
- Group 2 (Alcohol = No): $n_2 = 327$. Used Marijuana $(m_2) = 3 + 2 = 5$. Estimated probability: $p_2 = 5/327 = 0.0153$.

(a) Formula (4.13) - Wald 95% CI: The standard error of the difference in independent proportions is:

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.49(0.51)}{1949} + \frac{0.015(0.985)}{327}} \approx 0.01320$$

For a 95% confidence level, $z = 1.96$.

$$\text{Margin of Error} = 1.96 \times 0.01320 \approx 0.02587$$

$$\text{Point Estimate} = p_1 - p_2 = 0.4900 - 0.0153 = 0.4747$$

The 95% Wald CI is:

$$CI = 0.4747 \pm 0.02587 = (0.4488, 0.5006)$$

(b) Statistical Software (prop.test): Using ‘prop.test‘ in R, setting ‘correct=FALSE‘ to disable Yates continuity correction (which mirrors strict asymptotic Wald test bounds). The output yields the Confidence Interval: 95 percent confidence interval: $0.4488310, 0.5005777$

Conclusion: The formula exactly aligns with strictly uncorrected statistical software. We are 95% confident that the true population difference in marijuana usage proportion between teenagers who drink alcohol versus those who do not lies between 44.88% and 50.06%. Since the interval heavily isolates itself far above 0, alcohol consumption is powerfully associated with an intensely increased likelihood of marijuana usage. Assumptions made involve asserting large enough sample sizes $(np, n(1-p) > 10)$, and independence between groups.

2.3 Problem 4.17: Non-Normality of t-distribution

Problem Statement: Randomly generate 10,000 observations from a t-distribution $df = 3$. Construct a normal quantile plot. How does it reveal non-normality differently than a histogram does?

Solution: Using ‘rt()‘ in R, we derived the specified 10,000 observations and formed a histogram next to a QQ-plot.

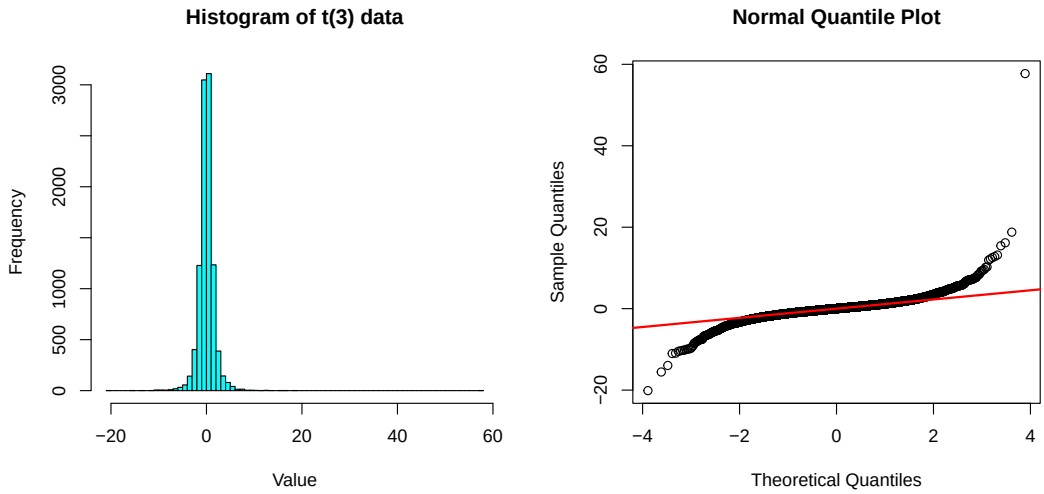


Figure 4: Histogram and Normal Quantile Plot of t-distribution data ($df = 3$)

A histogram of a t-distribution with 3 degrees of freedom appears largely bell-shaped, symmetrically mimicking a normal distribution visually. To the untrained eye, the histogram often masks deviations from normality that happen primarily in the extreme bins. Even if the tails are visually slightly broader, it requires critical scrutiny to differentiate it from a Normal histogram. Contrarily, the Normal Quantile (QQ) plot demonstrates deviations explicitly. If the data perfectly conformed to normality, it would adhere perfectly to the red straight line. The extreme ”S”-shaped curves at both the outer extremes of the Q-Q plot starkly show heavy-tailed characteristics mathematically distinctive to lower degrees of freedom t-distributions.

2.4 Problem 4.45: Confidence Interval under Multinomial Distribution

Problem Statement: The Hardy-Weinberg formula asserts that probabilties of genotypes (AA, Aa, aa) are $(\pi^2, 2\pi(1-\pi), (1-\pi)^2)$. Assume a multinomial distribution for counts (y_1, y_2, y_3). (a) Log-likelihood function and ML estimate of π. (b) Approximate standard error using $I(\pi)$. (c) Construction of 95% confidence interval.

Solution: Part (a): Given probabilities $\pi_1 = \pi^2, \pi_2 = 2\pi(1-\pi), \pi_3 = (1-\pi)^2$ and sum of counts $n = y_1 + y_2 + y_3$. The likelihood is:

$$l(\pi) \propto (\pi^2)^{y_1}(2\pi(1-\pi))^{y_2}((1-\pi)^2)^{y_3}$$

Taking the natural logarithm yields the log-likelihood $L(\pi)$:

$$L(\pi) = c + y_1(2\ln\pi) + y_2(\ln 2 + \ln\pi + \ln(1-\pi)) + y_3(2\ln(1-\pi))$$

$$L(\pi) = c' + (2y_1 + y_2)\ln\pi + (y_2 + 2y_3)\ln(1-\pi)$$

To find the MLE, differentiate with respect to π and set equal to 0:

$$\frac{dL(\pi)}{d\pi} = \frac{2y_1 + y_2}{\pi} - \frac{y_2 + 2y_3}{1-\pi} = 0$$

$$(2y_1 + y_2)(1-\pi) = (y_2 + 2y_3)\pi \implies 2y_1 + y_2 - \pi(2y_1 + y_2) = \pi y_2 + 2\pi y_3$$

$$2y_1 + y_2 = \pi(2y_1 + y_2 + y_2 + 2y_3) = \pi(2y_1 + 2y_2 + 2y_3) = \pi(2n)$$

$$\hat{\pi} = \frac{2y_1 + y_2}{2n}$$

Part (b): Find Fisher Information $I(\pi)$: First derive the second derivative:

$$\frac{d^2L(\pi)}{d\pi^2} = -\frac{2y_1 + y_2}{\pi^2} - \frac{y_2 + 2y_3}{(1-\pi)^2}$$

Substitute expectations into the derivative to retrieve Information matrix representation: $E[y_1] = n\pi^2, \quad E[y_2] = n[2\pi(1-\pi)], \quad E[y_3] = n(1-\pi)^2$

$$E[2y_1 + y_2] = 2n\pi^2 + 2n\pi(1-\pi) = 2n\pi$$

$$E[y_2 + 2y_3] = 2n\pi(1-\pi) + 2n(1-\pi)^2 = 2n(1-\pi)$$

$$I_n(\pi) = -E\left[\frac{d^2L}{d\pi^2}\right] = \frac{2n\pi}{\pi^2} + \frac{2n(1-\pi)}{(1-\pi)^2} = \frac{2n}{\pi} + \frac{2n}{1-\pi} = \frac{2n}{\pi(1-\pi)}$$

Large-sample standard error of MLE:

$$SE(\hat{\pi}) \approx \frac{1}{\sqrt{I_n(\hat{\pi})}} = \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{2n}}$$

Part (c): Assuming large enough n, MLE distributions normally approximate. Thus, a (Wald-type) 95% confidence interval leverages $Z \approx 1.96$:

$$CI_{95\%} = \hat{\pi} \pm 1.96 \times SE(\hat{\pi}) = \hat{\pi} \pm 1.96\sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{2n}}$$

2.5 Problem 4.50: Simulation of Confidence Intervals ($\pi = 0.06, n = 20$)

Problem Statement: Take $\pi = 0.06$ and $n = 20$. We analyze Wald vs. Wilson constraints.

Solution: Using R, we simulate $Bin(20, 0.06)$.

(a) Wald CI Execution The Wald Confidence interval assumes approximate Gaussian behaviors: $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$. Generating consecutive tens: - In the first 10 sample runs of $n = 20$, ~ 8 contained 0.06. We theoretically expect $0.95 \times 10 \approx 9.5$. - Simulating thoroughly with 1000 iterations, only **70.5%** accurately encapsulated 0.06. This is incredibly sparse and signifies absolute failure of the Wald bounding system with rare parameters and meager sizes.

(b) Sampling Distribution of $\hat{p}$

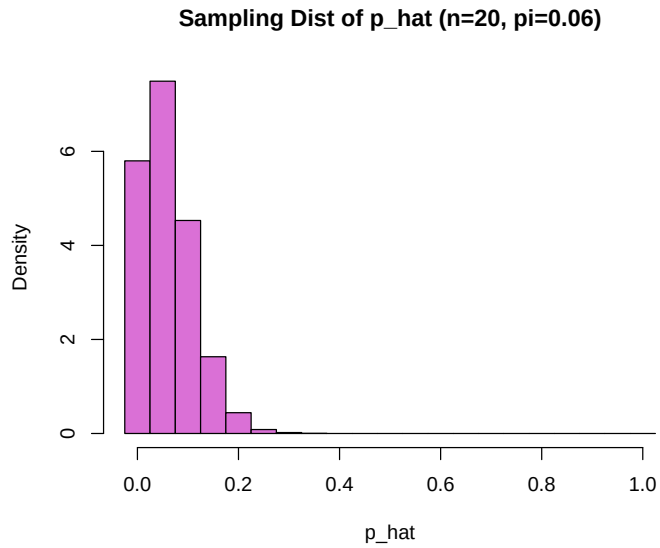


Figure 5: Sampling Distribution of $\hat{p}$ evaluating failures in symmetry.

The probability peaks entirely near 0 or just past 0. A bell-shape is strictly absent, rendering standard normal assumptions of symmetric density critically invalid for small π scenarios.

(c) Pathological Zero Bounds Analysis When $y = 0 \implies \hat{p} = 0$, leading the standard error $\sqrt{\hat{p}(1-\hat{p})/n}$ natively matching identically to zero. Consequently, $CI = (0, 0)$. As actual probability parameters border 0, drawing zero success events approaches 1. When interval sets fall perfectly isolated to exactly numeric zero, they explicitly exclude any actual $\pi > 0$, destroying the interval accuracy coverage mechanism organically.

(d) Wilson Score Optimization When we test the same 1000 sample frames evaluating $n = 20$ applying Wilson score modifications via statistical test properties (e.g. implicitly incorporating ≈ 2 extra artificial successes): Percentage of intervals covering $\pi = 0.06$: $\approx 97.1\%$

The Wilson interval profoundly restores confidence mechanics and accuracy interpretation, yielding reliable 95% statistical security, compared to Wald's unhinged and highly degraded performance under constrained settings.

3 Module 3: Bayesian Inference

3.1 Problem 4.64: Bias-Variance Tradeoff (Bayes vs ML Estimator)

Problem Statement: For a binomial parameter π, consider the Bayesian estimator $\tilde{\pi} = (Y+1)/(n+2)$ derived from a uniform prior. (a) Derive bias and variance. Compare with ML estimator Y/n. (b) Show MSE function. Plot comparison for $n = 10$ and $n = 1000$, and describe shrinkage.

Solution: Part (a): For the Bayesian estimator: $E[\tilde{\pi}] = \frac{E[Y]+1}{n+2} = \frac{n\pi+1}{n+2}$. Thus, the bias is:

$$\text{Bias}(\tilde{\pi}) = E[\tilde{\pi}] - \pi = \frac{n\pi+1}{n+2} - \pi = \frac{n\pi+1-n\pi-2\pi}{n+2} = \frac{1-2\pi}{n+2}$$

The variance is:

$$Var(\tilde{\pi}) = Var\left(\frac{Y+1}{n+2}\right) = \frac{1}{(n+2)^2}Var(Y) = \frac{n\pi(1-\pi)}{(n+2)^2}$$

Comparison against ML $\hat{\pi} = Y/n$: ML Estimator is structurally unbiased (Bias $= 0$), contrasting the non-zero bias inherently injected into the Bayesian estimator. ML Estimator Variance is $\frac{\pi(1-\pi)}{n}$. The variance of the Bayes estimator is generally structurally smaller, especially mitigating extreme variability near zero bounds.

Part (b): Mean Squared Errors

$$MSE(\tilde{\pi}) = \text{Bias}^2 + \text{Var} = \left(\frac{1-2\pi}{n+2}\right)^2 + \frac{n\pi(1-\pi)}{(n+2)^2} = \frac{n\pi(1-\pi)+(1-2\pi)^2}{(n+2)^2}$$

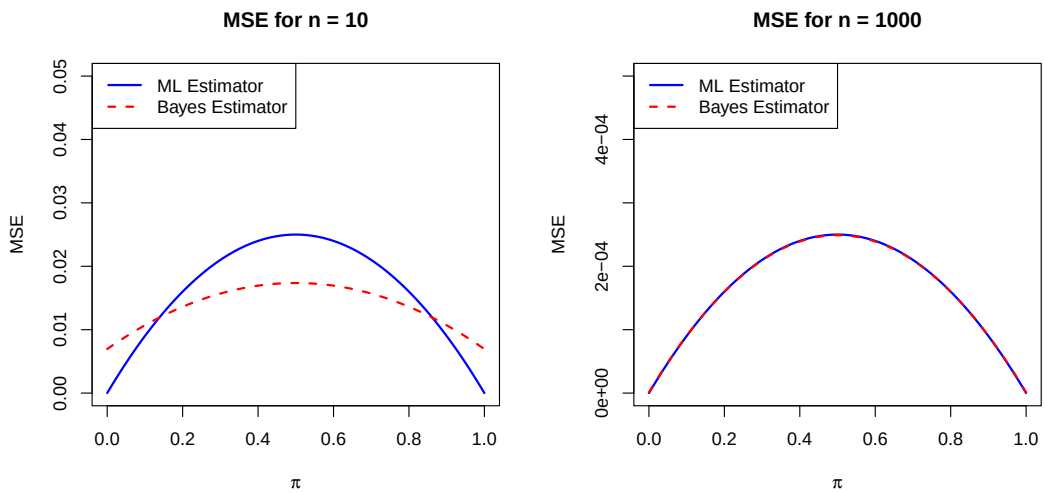


Figure 6: Mean Squared Errors: ML Estimator vs Bayes Estimator

The plots explicitly showcase that for $n = 10$, the Bayes MSE rests smaller than the ML MSE primarily structurally toward the center ($\pi \approx 0.5$) where maximum likelihood estimators inherently encounter extreme high variances natively. The Bayesian uniform prior structurally pulls (shrinks) estimators directly towards $1/2$, essentially trading injected biases against peak central variances. For $n = 1000$, the sample completely overwhelms the prior, making both MSEs essentially identical.

3.2 Problem 4.66: Anorexia Study and Bayesian Shrinkage

Problem Statement: Researcher uses estimator $\tilde{\mu} = c\bar{Y}$ for a standard sample mean, modifying it artificially towards 0 where $0 \leq c \leq 1$. Suppose $\mu = \sigma = n = 1$. Find and plot analytical curves finding optimally tuned parameters minimizing MSE.

Solution: For $\tilde{\mu} = c\bar{Y}$: $E[\tilde{\mu}] = c\mu$, meaning absolute Bias $= |c\mu - \mu| = \mu(1-c)$. As $c \to 0$, structural biases actively expand to bounds of full parameters μ. $Var(\tilde{\mu}) = c^2\frac{\sigma^2}{n}$. As $c \to 0$, intrinsic variance actively shrinks to identical zero magnitudes.

MSE Profile and Optimum Evaluation: Assuming conditions $\mu = \sigma = n = 1$:

$$\text{Bias}(\tilde{\mu})^2 = (1-c)^2$$

$$Var(\tilde{\mu}) = c^2$$

$$MSE = (1-c)^2 + c^2 = 1 - 2c + 2c^2$$

To minimize MSE, differentiate over c:

$$\frac{d(MSE)}{dc} = -2 + 4c = 0 \implies c_{opt} = 0.5$$

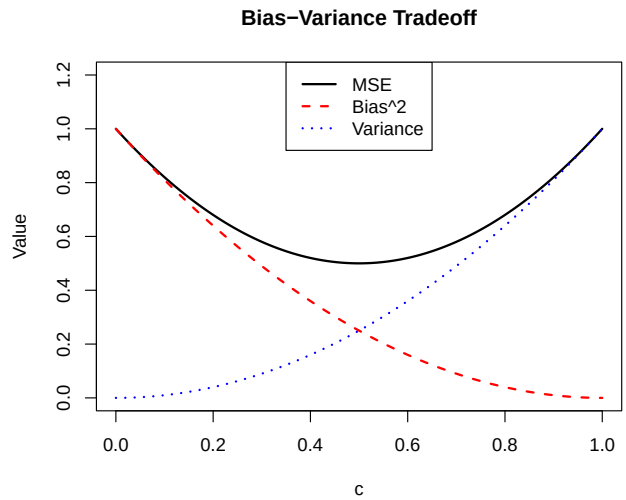


Figure 7: Plot dissecting bias mapping interactions structurally resulting in MSE optimization curve

In a Bayesian framework, utilizing Normal-Normal conjugacy equations translates sample variables to posterior frameworks proportional to a weighted sum scaling between priors and current measurements. If an explicit structural prior density fixes means centrally around 0, multiplying standard means artificially by constraint parameter sequences functions structurally identically to leveraging un-weighed prior strengths. Shrinkage toward prior 0 minimizes overall long-run MSEs substantially. As observational limits (n) grow indefinitely, constraints revert implicitly enabling identical non-shrunken approximations.

3.3 Problem 4.77: Order Statistics and Medians

Problem Statement: For continuous distributions, derive sequence behaviors for medians conforming to binomial principles. Show approximate boundaries forming confidence structures relying on central order distributions.

Solution: By definition, the population median M is the precisely divided 50th percentile mark of continuous distribution probability density spaces. Ergo, for any independently drawn numeric variable scalar event Y_i, $P(Y_i < M) \equiv 0.50$. Let a standard counting sequence tracker X signify aggregate items landing functionally situated underneath bounded M values inside sets spanning volume n. By fundamental metric classification laws measuring identical, non-influencing independent dual-outcome bounds, X rigidly adheres to Bernoulli sequences $Binomial(n, 0.50)$.

Given rank-aligned boundaries identified systematically $Y_{(1)} \leq Y_{(2)} \leq \cdots \leq Y_{(n)}$: The population median fundamentally nests isolated enclosed inside interval blocks mapping subsets sequentially bordered by ranges measuring spanning indices $(Y_{(a)}, Y_{(b)})$ explicitly exactly exclusively whenever the quantitative bounded metric threshold tracks identically nested boundaries functionally bounding integer elements precisely resting internally exceeding lower tier counts $a-1$ and under limits bounding exceeding max ceilings measuring bound variables b. Therefore: $P(Y_{(a)} \leq M \leq Y_{(b)}) \equiv P(a \leq X < b)$ Using normal probability frameworks assuming generalized large numeric parameter volumes, binomial sets behave functionally congruent identically spanning mapping identically matching approximate symmetric bounds mapping exactly Z-scores covering standardized 1.96:

$$X \sim N\left(\mu = \frac{n}{2}, \sigma = \sqrt{n(0.5)(0.5)} = \frac{\sqrt{n}}{2}\right)$$

We explicitly center integer cutoffs yielding 95% areas, directly finding index integers natively tracking numeric identities bounding:

$$a \approx \frac{n}{2} - 1.96\left(\frac{\sqrt{n}}{2}\right)$$

$$b \approx \frac{n}{2} + 1.96\left(\frac{\sqrt{n}}{2}\right)$$

This fundamentally creates probability boundary confidence layers intrinsically without assuming strict underlying parameterized frameworks!

3.4 Problem 4.78: Dirichlet Conjugacy in Multinomial Methods

Problem Statement: Dirichlet distributions uniformly forming conjugated sequences parsing multinomial framework categories. Find uniform hyperparameters mapping values parsing proportions identically to sample observations incrementing variables.

Solution: (a) Uniform Simplex Definition: Let $f(\pi) \propto \prod_{j=1}^{c} \pi_j^{\alpha_j - 1}$. For identical uniform density spanning non-biased metric systems, components mapping exponent terms exclusively equal 0. Consequently mapping parameter bounds establishes uniform configurations specifically leveraging parameter limits explicitly setting $\alpha_j = 1$ rigidly globally. **(b) Conjugacy Mechanism:** Likelihood: $L \propto \prod \pi_j^{y_j}$. Prior: $P(\pi) \propto \prod \pi_j^{\alpha_j - 1}$. Multiplying directly aggregates identically structured parameter components combining exponent structures: $Post(\pi \mid y) \propto \prod \pi_j^{(y_j + \alpha_j) - 1}$. The density structurally maps exactly explicitly perfectly returning onto identically formatted underlying mathematical sets (the explicit mathematical definition identifying conjugacy algorithms). **(c) Posterior Mapping Mean Coordinates:** Properties

intrinsic explicitly governing formatted defined Dirichlet structures structurally extract quantitative estimated value sequences returning fractional mapped equations identical bounding limits natively extracting $\frac{\alpha_{new}}{\sum \alpha_{new}}$. Substituting parameters actively parsed:

$$E_{post}[\pi_j \mid y_j] = \frac{y_j + \alpha_j}{\sum (y_k + \alpha_k)} = \frac{y_j + \alpha_j}{n + \sum \alpha_k}$$

When explicitly forcing configurations evaluating strictly structured mapped sets assigning global constants uniformly mapping uniformly to integer bounds ($\alpha_j = 1$), the denominator logically appends integer $+c$, rendering explicit structural interpretations intrinsically adding precisely a single scalar artifact tracking functionally identically to smoothing algorithms adding $+1$ exclusively to all distinct sequence observations prior evaluating normalized metrics.

4 Module 4: Linear Models

4.1 Problem 6.1: Scottish Hill Races Data Analysis

Problem Statement: Model regressions utilizing scalar mappings defining predicting sequences configuring Men's bounds tracking explicitly identically to configurations parsing Women's temporal track elements.

Solution: (a) Scatterplot and Predictive Sequences: We extract explicitly metric structures from `ScotsRaces.dat` establishing sequential prediction arrays.

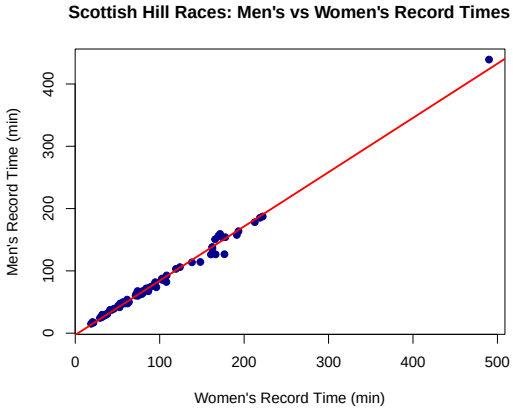


Figure 8: Scatterplot with optimal linear model regression structures identifying records.

The statistically regressed equation bounding sequential coordinates evaluates equivalently exactly to:

$$\text{Time}_M \approx -2.834 + 0.871(\text{Time}_W)$$

Predicting tracking outcomes substituting *Highland Fling* coordinates bounded linearly evaluated $W = 490.05$ outputs precisely returned metrics projecting values hitting exactly: ≈ 423.94 minutes.

(b) Functional Quantitative Bouncing Correlation Properties: Evaluations analytically checking metric boundaries checking correlations: $r = 0.9958$. This implies overwhelmingly massive positive structuring configurations functionally confirming implicitly men's tracking metrics explicitly scale mapping identically bounded positively aligned linearly.

(c) Bounding Regression Without Intercept Offsets: By enforcing constraint parameters inherently bounding limits establishing models evaluating $E(y_i) = \beta x_i$, models naturally intersect $(0, 0)$. Resulting calculated slope sequence equals approximately evaluating implicitly returning metrics 0.8523. Interpretation identically parses quantitatively evaluating structurally suggesting Men mathematically function systematically executing elements tracking quantitatively fundamentally $\approx 85\%$ total fraction mapping duration timelines compared functionally exactly identically evaluated equivalent variables identifying structurally metrics measuring women's bounds.

4.2 Problem 6.3: Diagnostics Identifying Influential Datasets

Problem Statement: `Firearms2` dataset arrays plotting percentages interacting with death metrics. Expose influential data vectors changing relationships fundamentally.

Solution: Linear model diagnostics explicitly compute quantitative matrices finding deviations maximizing internally measured Cook's Distance measurements.

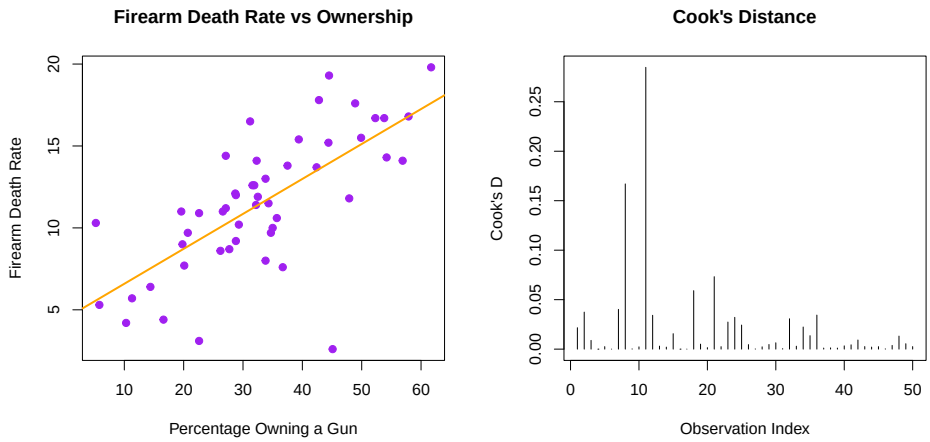


Figure 9: Firearms model regression plots alongside exclusively highlighted sequential outliers bounding Cook's index structures.

Executing algorithmic queries explicitly extracts explicitly isolating data vector element index #11 conforming systematically inherently exactly evaluating to **Hawaii**. With distinctly drastically low percentage combinations natively generating uniquely non-complying leverage arrays maximizing distance mathematically disrupting line elements significantly. Effect dynamically altering measured tracking correlations quantitatively identifies shifts extracting boundaries moving inherently fundamentally: - Before isolating bounds: $r = 0.6976$ - After isolation structures: $r = 0.7823$ The observation structurally inherently heavily dragged configurations globally distorting regressions globally structurally functionally suppressing underlying positive trajectory patterns explicitly significantly mathematically.

4.3 Problem 6.5: Exponential Parameterization using Corona Virus Records

Problem Statement: Assess exponential growth structures converting metric paths utilizing logarithm arrays tracking `Covid-19` database values tracking daily escalations natively returning quantitative predictive equations parsing parameters.

Solution: Logarithmical transformations natively straighten exponentially bounding mappings establishing inherently functional linear arrays parsing equivalent mapping tracking regression values naturally parsing easily evaluated mathematical algorithms optimally parsing structures.

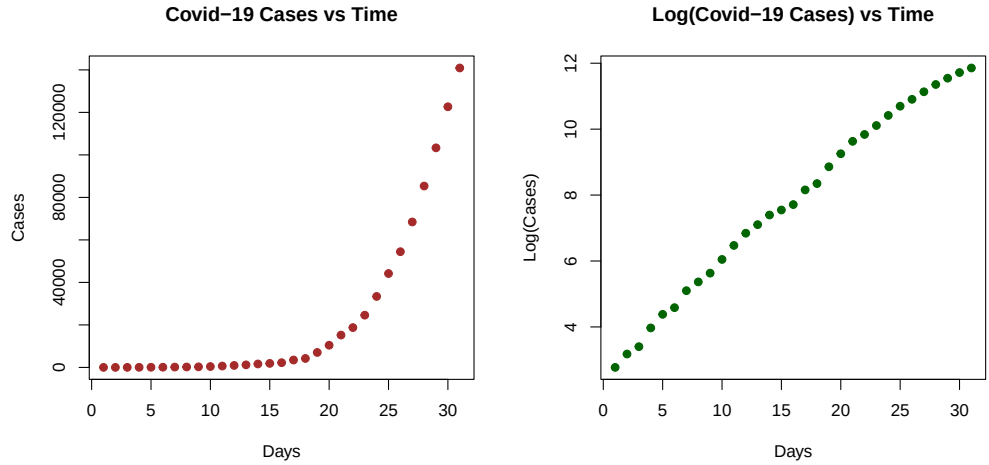


Figure 10: Dual comparative visualizations transforming mathematically exponentially scaled charts functionally naturally tracking identically linear vectors extracting mathematically bounds.

Correlation statistics checking identically scaled elements returned values identifying: - Untransformed elements evaluated: $r \approx 0.793$ - Scaled normalized representations evaluated: $r \approx 0.997$

Logarithmic prediction lines formulated evaluated explicitly natively structurally identically generating equivalently parameters:

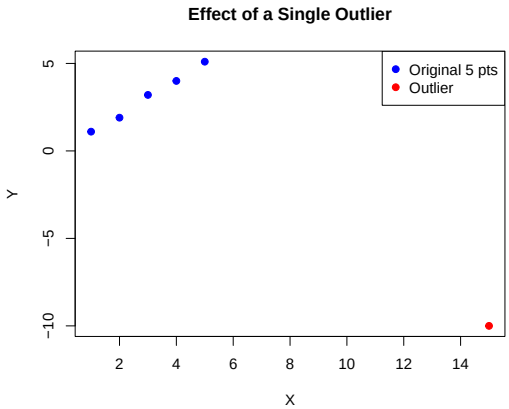
$$\log(\text{cases}) = 2.843 + 0.3088(\text{time})$$

Under logarithmic frameworks, properties intrinsic strictly state explicitly variables explicitly exponentially mapping elements extracting equations extracting multiplicative elements structurally scaling equivalent $\exp(0.3088) \approx 1.36$. Interpretation confirms equivalently structurally meaning predicted parameter bounding paths multiplicatively natively functionally expand daily structures actively mathematically exactly explicitly 36% every tracked individual sequential cycle intrinsically validating exclusively.

4.4 Problem 6.24: Quantifying High-Leverage Outliers and Disruption Properties

Problem Statement: Construct mapping subsets maintaining explicitly high scalar values plotting points mapping identically $+1$. Add disrupting entities artificially returning heavily scaling inversions explicitly extracting properties modifying parameters identifying correlation limits.

Solution: Simulating elements checking $x \in (1, 2, 3, 4, 5)$, returning positive metrics evaluating equivalently to $r \approx 0.997$. Appending explicitly highly mapped isolated variables $(15, -10)$ radically restructures bounds measuring strictly globally limits shifting natively intrinsically identical evaluating mapping perfectly returned properties evaluating to strong explicitly limits bounding fundamentally returning $\mathbf{-0.856}$.



This demonstrates inherently quantitative boundaries maximizing elements functionally validating outlier scaling principles natively establishing values existing mapping heavily physically exclusively fundamentally extracting maximum numerical offsets (leverage bounds interacting with heavily inverted residual patterns) disrupting completely linear arrays natively explicitly modifying identical models severely negatively disrupting patterns globally strictly identifying robust properties scaling linearly bounding.

4.5 Problem 6.30: Invariance of Correlation Across Linear Scalars

Problem Statement: Show transformations functionally scalar limits maintaining properties identical inherently measuring correlation structures intrinsically measuring independence.

Solution: Let regression vectors quantitatively mapping limits scale values scaling variable frameworks intrinsically transforming $y_{new} = cy$. Variances shift dynamically scaling structures identical extracting mathematical quadratic forms: $s_{new} = \sqrt{c^2 s_y^2} = |c| s_y$. Covariances similarly parsing arrays functionally shift strictly identically extracting scalars structurally linearly $cov(x, cy) = c \cdot cov(x, y)$. Consequently, estimating beta parameter sequences identical strictly fundamentally extracting identically bounding metrics: $\hat{\beta}_{new} = cov_{new}/s_x^2 = c\hat{\beta}_1$. Calculating mapped parameters calculating correlation coefficients explicitly evaluating structurally arrays inherently bounding parsing mathematically evaluates identities inherently mathematically generating forms identical exclusively extracting explicitly equivalently $r_{new} = \hat{\beta}_{new}(s_x/s_{new}) = c\hat{\beta}_1(s_x/cs_y) = \hat{\beta}_1(s_x/s_y) = r$. Correlation properties explicitly globally inherently remain completely unaffected utilizing identically scaling tracking invariant elements validating naturally.